AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Final Examination
Fall Semester 2024/2025

13 /Jan. /2025

umber of Exam Pages: 3 cluding this cover sheet)	**Time Allowed: 2:00 Hours**

nstructions:

- Total Marks: **50 marks**
- This exam consists of 5 questions
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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You should answer all the questions, giving 50 possible marks. Show your work. Type equation here.

Q1. [3+2+2+ 3 Marks]

a) Given the function f(x) -5 = $-(x+2)^2$

i) Find the vertex and the axis of symmetry.

ii) Determine the minimum or maximum value of f(x). Justify your answer.

iii) Find the zeros for the quadratic function f(x).

b) Use synthetic division to divide $x^5+x^4+x^3+x^2$+x+1 by x+1

Q2. [3+4+3 marks]

a) Find the equation of the line passes through the point (5, 4) and is parallel to the line with equation 6 x+3y-4 =0

b) Find the inverse of the function f(x)=$\sqrt[3]{3x-5}$

c) The variable y is directly proportional to x-2 and y=20 when x =6, Write an equation-relating x and y and find y when x= 12

Q3. [4+2+4marks]

Solve for x:

a) $\frac{3x+2}{2x-7} < 0$

b) Find the domain of the following and write the answer in interval notation

i) F(x)= $\sqrt{x^2-9}$

ii) G(x)= $\frac{x+1}{x^3-9x}$

Q4: [3+3+4 marks]

Solve for x:

a) $e^{x^2-2x} = e^8$

b) $\ln(\ln(3x)) = 0$

c) $\ln(x+1) = \ln(x-2)+1$

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Q5. [2+2+3+3marks]

a) How many distinct permutations can be made from the letters of the word COLUMNS?

b) A random sample of 200 adults are classified below by sex and their level of education attained.

Education	Male	Female
Elementary	38	45
secondary	28	50
College	22	17

If a person is picked at random from this group, find the probability that:

i. **The person is a male and has secondary education.**

ii. **The person is a male or has secondary education**

iii. **The person is a male, given thar the person has a secondary education.**

End of questions

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